

Literature Organization for IEEE Access Paper — TN-NTN Integrated Simulator (OMNeT++ / Simu5G / INET / VEINS / SUMO)

Purpose of this document: organize your 244-entry `export.bib` library into the Background section of your paper, positioned around your core contribution claim — *"no existing open, complete TN-NTN integrated simulator combining OMNeT++ (Simu5G, INET) with SUMO (via VEINS) for heterogeneous UE mobility, released open source for academia."* Papers are sorted into four buckets (Overview / Technical Contribution / Theory-New Perspective / Simulator Built), then by theme. Author-importance buckets, competitive-landscape gap analysis, additional SOTA found via web search, and a standards-body reference list follow.

Note on library composition: ~60 of the 244 entries are legacy Iridium/Globalstar system reports, unrelated UAV-robotics/SLAM papers, or malformed placeholder entries (bare conference proceedings, duplicate/garbled titles). These are listed at the end under "Exclude / marginal" rather than repeated throughout — recommend cleaning these out of Zotero/Mendeley before final BibTeX export.

1. How the four buckets map to your Background section

Bucket	What belongs here	Role in your paper
Overview	Surveys, tutorials, roadmaps, standardization reviews	Establish the field, cite once per sub-topic, don't over-cite
Technical Contribution	Papers proposing a specific algorithm/protocol/architecture (handover, beamforming, routing, edge computing, PHY/MAC)	Show breadth of what <i>other</i> researchers study analytically/numerically — motivates <i>why a shared simulator matters</i> (everyone builds one-off tools)
Theory / New Perspective	Analytical/stochastic-geometry models, AoI, new conceptual framings	Show the theoretical grounding your simulator could help validate empirically
Simulator Built	Papers that <i>build and release/describe a simulation tool or testbed</i>	Your direct competitive landscape — the most important bucket for justifying novelty

2. SIMULATOR BUILT — your direct competitive landscape (read these first)

This is the set you must position against explicitly in your Related Work / gap analysis.
Sorted chronologically.

Year	Title	Authors	Tool / Stack	Relevance to your gap claim
2011	OMNeT++ and MiXiM-based protocol simulator for satellite network	Li, Wang, Liu, Hu, Xu, Chen	OMNeT++/MiXiM	Early OMNeT++ satellite work — no 5G/terrestrial integration, no SUMO
2014	Satellite model for Network Simulator 3 (SNS3)	Puttonen, Rantanen, Laakso et al. (Magister Solutions) — <i>found via search, not in your .bib, add it</i>	ns-3	GEO/DVB-S2/RCS2 focus, not cellular NTN
2019	A Simulator for Creating Drones Networks and Providing Users Connectivity	Tropea, Fazio	Custom FANET sim	Drone-only, no satellite/cellular
2020	Simu5G: A system-level simulator for 5G networks	Nardini, Stea, Virdis, Sabella	OMNeT++/INET	Foundational tool you build on — no NTN/satellite support natively
2020	SiMu5G — An OMNeT++ library for end-to-end 5G evaluation	Nardini, Sabella, Stea, Thakkar, Virdis	OMNeT++/INET	Companion/duplicate of above — cite once
2021	Satellite integration into 5G: testbed for 5G Aero backhauling	Goratti, Herle, Betz et al.	Physical testbed (Starlink)	Real hardware, not simulation — good contrast point
2021	Emergency 5G Communication on-the-Move (satellite backhaul field trial)	Völk, Schwarz, Lorenz, Knopp	Field trial, EPC core-edge split	Real trial, not a released simulator
2022	Field Trial of a 5G NTN Using OpenAirInterface	Volk, Schlichter, Kaltenberger, Heyn, Casati,	OAI (real HW/GEO)	OAI = SDR-based, not discrete-event; huge cost/complexity vs. your OMNeT++ approach

Year	Title	Authors	Tool / Stack	Relevance to your gap claim
		Schwarz, Knopp		
2022	OpenAirInterface as a platform for 5G-NTN Research and Experimentation	Kumar, Meshram, Astro, Querol, Schlichter, Casati, Heyn, Volk, Schwarz, Knopp, Marques, Pereira, Magueta, Kapovits, Kaltenberger	OAI	Same OAI ecosystem — cite as the "SDR/testbed alternative" to pure simulation
2023	Experimental Study of RLC Modes for 5G-NTN using OpenAirInterface5G	Kumar, Sheemar, Querol, Nik, Chatzinotas	OAI + channel emulator	GEO only, no mobility/road-traffic integration
2023	Multipath Transport Analysis Over Cellular and LEO Access for Aerial Vehicles	Baltaci, Chavali, Kosek, Mohan, Schupke, Ott	Custom MP emulation testbed	Emulation, not full network-level sim
2024	A 5G NTN Emulation Platform for VNF Orchestration	Muro, Baena, De Cola , Fortes, Barco	Custom emulation platform	VNF/orchestration focus, not open discrete-event network sim
2024	Integrating Terrestrial and Non-Terrestrial Networks via IAB Technology: System-Level Design and Evaluation	Pugliese, Quadrini, Striccoli, Roseti, Zampognaro, Piro, Grieco, Boggia	Custom system-level simulator	Closest prior "TN+NTN system-level simulator" — but IAB-specific, not OMNeT++/Simu5G/SUMO based; cite prominently as closest competing tool

Year	Title	Authors	Tool / Stack	Relevance to your gap claim
2024	LEON: Simulating Handover Integrating Non-Terrestrial Networks with 5G and beyond	Zhang, Zhong, Yang, Schulz, Nguyen, Fitzek	OMNeT++ + INET + Simu5G + OS3	Nearly identical stack to yours — must be discussed explicitly; differentiate on (a) no SUMO/VEINS heterogeneous UE mobility, (b) handover-only scope vs. your general TN-NTN framework
2026	Design and implementation of 3GPP NTN protocol simulation platform	Tao, Sun, Yang, Sun, Cao	ns-3	Competing ns-3-based NTN stack, RRC per 3GPP TS 38.331 — no terrestrial road-traffic integration
2026	<i>A Disaster-Aware Integrated TN-NTN System-Level Simulator for Resilient 6G Wireless Networks</i> — found via web search, not in your .bib, add it	(arXiv 2605.05852)	OMNeT++-based TN/NTN split simulator	Very close to your contribution — TN/NTN dynamic switching, urban scenario, Germany-based — must review and differentiate against

Additional simulator/tool papers found via search — not currently in your library, recommended additions

Title	Venue/Year	Stack	Why it matters for your gap claim
Jiang, Zhan, Xiao et al., "Network Simulators for Satellite-Terrestrial Integrated Networks: A Survey"	IEEE Access, 2023	Survey of ~20+ simulators	The single most important paper to cite — it is the canonical survey your "no complete TN-NTN simulator" claim must engage with directly; confirm your gap still holds against its taxonomy

Title	Venue/Year	Stack	Why it matters for your gap claim
Kassing, Bhattacharjee, Águas et al., "Exploring the Internet from Space with Hypatia"	ACM IMC 2020	ns-3 + custom	Widely cited LEO constellation simulator (routing/latency), no terrestrial cellular integration
Zhuo et al./various, "StarryNet: Empowering Researchers to Evaluate Futuristic Integrated Space and Terrestrial Networks"	USENIX NSDI 2023	Container-based emulation	Direct "integrated space+terrestrial" claim — must be compared/contrasted
"SNS3 — Satellite Network Simulator 3" (Magister Solutions) and its 5G-NTN extension	ns-3	DVB-S2/RCS2, later 5G NTN	Commercial-academic ns-3 satellite line, relevant prior art
"5G-LENA NTN extension" (CTTC)	ns-3	5G NR NTN handover, BWP, CC config	Used within 3GPP standardization studies — important comparator
Puddu, Popescu, Murroni, "An Open Source Simulator for Next Generation Satellite Broadband Traffic Management"	IEEE Aerospace Conf. 2022	Custom	Traffic management focus, GEO
"Plotinus: A Satellite Internet Digital Twin System"	2024 (arXiv 2403.08515)	Digital twin	Positions against Hypatia/StarryNet/LEOCN — useful for your related-work table
"LEOCN: Real-time and complete network simulation framework for LEO constellation networks"	J. High Speed Networks 2024	Custom	Another LEO-only competitor, no terrestrial UE/road integration

Title	Venue/Year	Stack	Why it matters for your gap claim
"Network Simulator for Large Low Earth Orbit Satellite Networks" (survey, comprehensive review of ns-3-based LEO frameworks)	~2021-2024	Survey	Companion survey to Jiang et al. — cite alongside it
5GSimV2IN (GitHub: cs7org/5GSimV2IN)	OMNeT++ + Simu5G + INET + VEINS + SUMO	Directly overlapping toolchain with yours (V2X/V2N, not NTN) — critical to cite: shows the SUMO+VEINS+Simu5G integration pattern exists for <i>terrestrial</i> V2X but has never been extended to NTN — this is your precise novelty wedge	
Deutschmann, Hielscher, German, "An ns-3 model for multipath communication with terrestrial and satellite links"	MMB 2023	ns-3	Terrestrial+satellite multipath, not full TN-NTN handover/mobility sim

Bottom line for your novelty framing: Your "no complete open-source TN-NTN simulator with heterogeneous UE mobility via SUMO/VEINS" claim is **still defensible**, but it must explicitly address and differentiate from: (1) LEON (closest OMNeT++/Simu5G/INET/OS3 stack — lacks SUMO/VEINS and multi-UE-type mobility), (2) the arXiv Disaster-Aware TN-NTN simulator, (3) Pugliese et al.'s IAB system-level simulator, (4) the Jiang et al. 2023 IEEE Access survey (must show your tool fills an identified gap in their taxonomy), and (5) 5GSimV2IN (proves the SUMO+VEINS+Simu5G recipe works for terrestrial V2X — you are the first to extend it to NTN).

3. OVERVIEW — surveys, tutorials, standardization reviews

Sort for Background section (chronological within sub-theme):

NTN/5G-6G standardization & architecture overviews

- Guidotti, Vanelli-Coralli et al., "Architectures and key technical challenges for 5G systems incorporating satellites" (2019, IEEE TVT)
- Guidotti, Cioni, Colavolpe, Conti, Foggi, Mengali, Montorsi, Piemontese, Vanelli-Coralli, "Architectures, standardisation, and procedures for 5G Satellite Communications: A survey" (2020, Computer Networks) — **cornerstone survey**
- Zhang, Zhu, Wang, "A survey on space-aerial-terrestrial integrated 5G networks" (2020, Computer Networks)
- Kim, "5G wireless communication technology for non-terrestrial networks" (2019, ETRI)
- Hosseinian, Choi, Chang, "Review of 5G NTN Standards Development and Technical Challenges" (2021)
- Sun, Peng, Zhang, Lin, Zhang, "Integrated Satellite-Terrestrial Networks: Architectures, Key Techniques, and Experimental Progress" (2022, IEEE Network)
- Guidotti, Vanelli-Coralli, Schena, Chuberre, El Jaafari, Puttonen, Cioni, "The path to 5G-Advanced and 6G Non-Terrestrial Network systems" (2022)
- El Jaafari, Chuberre, Anjuere, Combelles, "Introduction to the 3GPP-defined NTN standard" (2023) — **essential 3GPP-context reference**
- Masini, Reininger, El Jaafari, Vesely, Chuberre, Baudry, Houssin, "5G meets satellite: Non-terrestrial network architecture and 3GPP" (2023)
- Han, Dong, Retana, Li, Wang, Jiang, "Evolution to 6G for Satellite NTN Integration: From Networking Perspective" (2023, ICNP) — reviews IETF+3GPP jointly
- Westphal, Han, Li, "LEO Satellite Networking Relaunched: Survey and Current Research Challenges" (2023, arXiv)
- Guidotti, Vanelli-Coralli, El Jaafari, Chuberre, Puttonen, Schena, Rinelli, Cioni, "Role and Evolution of Non-Terrestrial Networks Toward 6G Systems" (2024)
- Geraci, Lopez-Perez, Benzaghta, Chatzinotas, "Integrating Terrestrial and Non-Terrestrial Networks: 3D Opportunities and Challenges" (2023, IEEE Comm. Mag.)
- Yao, Wang, Wang, Lu, Liu, "The Space-Terrestrial Integrated Network: An Overview" (2018)
- Chen, Lin, Lee, Toskala, Sun, Chiasserini, Liu, "5G-Advanced Toward 6G: Past, Present, and Future" (2023)
- Ye, Lo, Jeon, Tarver, Tonnemacher, Yeo, Cho, Xu, Kim, Zhang, "5G New Radio and Non-Terrestrial Networks: Reaching New Heights" (2022)
- Wang, Guo, Nauman, Qi, Wu, "Satellite Air Ground Integrated Networks: Architecture, Technologies, and Future Directions" (2026)

Handover / mobility surveys

- Chowdhury, Atiquzzaman, "Handover Schemes in Satellite Networks: State-of-the-Art and Future Research Directions" (2006) — classic taxonomy, still widely cited
- Chowdhury, Atiquzzaman, Ivancic, "Handover Schemes in Space Networks: Classification and Performance Comparison" (2006)

- Park, Kim, "Trends in LEO Satellite Handover Algorithms" (2021)
- Sridhar, Naik, "Optimizing QoS in LEO Satellite Networks: A Comprehensive Review of Handoff Algorithms" (2023)

Simulator-specific surveys (add from search)

- Jiang, Zhan, Xiao et al., "Network Simulators for Satellite-Terrestrial Integrated Networks: A Survey" (IEEE Access, 2023) — **must-cite**
- "A Comprehensive Review of ns-3-Based Simulation Frameworks for LEO Satellite Constellations" (add via search)
- "Network-Layer Perspectives on Satellite-Terrestrial Integrated Networks in 6G: A Comprehensive Review" (2025, Engineering/ScienceDirect)

Channel/PHY/link-budget surveys

- Zhu, Hofmann, Knopp, "Ka-Band LEO Satellite IoT Channel Characterization: Survey and Measurement" (2022)
- Giggenbach, Knopp, Fuchs, "Link budget calculation in optical LEO satellite downlinks" (2023)

Other domain surveys

- Rekkas, Sotiroudis, Sarigiannidis, Wan, Karagiannidis, Goudos, "Machine learning in beyond 5G/6G networks — state-of-the-art and future trends" (2021)
- Garcia, Molina-Galan, Boban, Gozalvez, Coll-Perales, Sahin, Kousaridas, "A Tutorial on 5G NR V2X Communications" (2021) — relevant since you use VEINS for road-side UE

4. TECHNICAL CONTRIBUTION — the bulk of the corpus

Organize your Background/Related Work by these sub-themes (each is a citable "cluster" — cite 3–6 representative papers per sub-theme, not all):

4a. NTN handover management (39 papers — largest cluster)

Split explicitly into two eras for narrative clarity:

Legacy LEO-MSS/ATM-era handover (1997–2015) — useful only as historical grounding, cite 2–3 max: Uzunalioglu/Yen/Akyildiz (1997) · Del Re/Fantacci/Giambene (1999, two papers) · Cho/Akyildiz/Bender (2002) · Papapetrou/Karapantazis/Dimitriadis/Pavlidou (2004) · Wang/Li/Liu/Hu (2013)

Modern 5G/6G-NTN handover (2016–2024) — cite more heavily, these are your closest technical peers:

- Wu, Jin, Luo, Fu, Shan, Hu, "A Graph-Based Satellite Handover Framework" (2016)

- Zhang, Liu, Han, Ding, Liang, "A Network-Flows-Based Satellite Handover Strategy" (2021)
- Kong, Lu, Yin, "Achieving Efficient and Secure Handover in LEO Constellation-Assisted B5G Networks" (2022)
- **Juan, Lauridsen, Wigard, Mogensen, "Handover Solutions for 5G Low-Earth Orbit Satellite Networks" (2022, IEEE Access)** and companion "Performance Evaluation of the 5G NR Conditional Handover in LEO-based NTN" (2022) — both from Nokia Bell Labs group, system-level-simulation-based, **strong methodological peer for your paper**
- Zhou, Liu, Zhang, Ouyang, Huang, "A Novel Feeder Link Handover Strategy for Backhaul in LEO Satellite Networks" (2023)
- Wang, Jiang, Wang, Chen, Quek, "Seamless Handover in LEO Based Non-Terrestrial Networks" (2023)
- **Lee, Park, Park, Molisch, "Handover Protocol Learning for LEO Satellite Networks: Access Delay and Collision Minimization (DHO)" (2024, IEEE TWC)** — DRL-based, strong recent benchmark
- Zhao, Wang, Zhao, Tarchi, Wan, Kumar, "INTERLINK: A Digital Twin-Assisted Storage Strategy for Satellite-Terrestrial Networks" (2022)

4b. Beamforming / MIMO / PHY for NTN (23 papers)

Dominated by the **Guidotti-Vanelli-Coralli-Riviello-Dakkak (Bologna)** and **Knopp-Delamotte-Tuzi (TU Munich)** groups:

- Ahmad, Riviello, Guidotti, Vanelli-Coralli — three related papers on graph-based user scheduling for LEO-MIMO NTN (2023, x3)
- Dakkak, Riviello, Guidotti, Vanelli-Coralli — MU-MIMO/federated beamforming for B5G/6G LEO NTN (2023, 2024, x2)
- Guidotti, Vanelli-Coralli, Amatetti, "Federated Cell-Free MIMO in Nonterrestrial Networks" (2024, IEEE TAES)
- Tuzi, Delamotte, Knopp et al. — satellite swarm/D2C antenna array papers (2023, x3)
- Cheporniuk, Schwarz, Delamotte, Knopp, "MIMO Throughput Performance Analysis in LEO Communication Scenario" (2021)
- Wünsche, Krondorf, Knopp, "Investigations of Channel Capacity Loss in LEO Satellite Systems Using Phased Array Beamforming Antennas" (2024)
- Lin, Lin, Zhu, Chatzinotas et al. — secure/RSMA beamforming for cognitive satellite-terrestrial networks (2018–2021, x4)

4c. Routing & inter-satellite-link (ISL) topology (20 papers)

- Hauri, Bhattacharjee, Grossmann, Singla, "Internet from Space without Inter-satellite Links?" (2020, ACM CoNEXT) — influential ISL-value analysis

- Li, Wu, Kang, Chen, Qiu, Zhou, "A Flexible Topology Control Strategy for Mega-Constellations via ISL" (2024)
- Feng, Sun, Peng, "Distributed Satellite-Terrestrial Cooperative Routing Strategy" (2024)
- Wu, Han, Chen, Wang, Meng, Benslimane, "Enhancing LEO Mega-Constellations with ISLs: Vision and Challenges" (2024)
- Rinaldi, Calatafimi, Pizzi, Molinaro, Araniti, "Delivering Multimedia Services in LEO satellite NTN Supported by Backward/Frontward ISL during Feeder Link Switch-Over" (2023)
- Zhang, Zhao, Li, Fang, "Fast Recovery from Multiple Link Failures in LEO Satellite Networks" (2023)
- Ashwood-Smith, McCormick, "Orthodromic Routing and Forwarding for Large Satellite Constellations" (2025)

4d. TN-NTN integration architectures (25 papers)

- Rinaldi, Pizzi, Molinaro, Iera, Araniti, "Cooperative Resource Allocation in Integrated Terrestrial/Non-Terrestrial 5G and beyond Networks" (2020)
- Kim, Casati, Cassiau, Pietrabissa, Giuseppi, et al., "Design of cellular, satellite, and integrated systems for 5G and beyond (5G-ALLSTAR)" (2020)
- Campana, Amatetti, Vanelli-Coralli, "RAN Functional Splits in NTN: Architectures and Challenges" (2023)
- Han, Xu, Liu, Yu, Zhao, Yu, "A Novel Mobile Core Network Architecture for Satellite-Terrestrial Integrated Network" (2021)
- Kwon, Shin, Conti, Lindsey, Win, "Access-Backhaul Strategy via gNB Cooperation for Integrated Terrestrial-Satellite Networks" (2024)
- Heine, Völk, Schwarz, Knopp, "Concept and Evaluation of 5G Backhauling via Starlink" (2022) — real Starlink measurements, good empirical anchor
- Zaglauer, Pfeifle, Gavras, Kapovits, Corici et al., "Integrated satellite access to 5G/6G systems — 5G space infrastructure study" (2022, ESA study)

4e. Edge computing / MEC in NTN (6 papers)

- Traspadini, Giordani, Giambene, De Cola, Zorzi, "On the Energy Consumption of UAV Edge Computing in NTN" (2023)
- Gao, Yan, Zhao, De Cola, Li, "Joint Optimization of Server and Service Selection in Satellite-Terrestrial Integrated Edge Computing Networks" (2024)
- Jia, Wu, Guo, Yang, "Service-Oriented SAGIN With Pervasive Intelligence for Resource-Constrained Users" (2024)

4f. UAV/HAPS aerial-network integration (24 papers)

- Sundaresan, Chai, Chakraborty, Rangarajan — SkyRAN/SkyLiTE/SkyHaul/SkyCore series (2018–2021, NEC Labs) — self-organizing UAV LTE RAN, core-to-edge —

strong systems-building peer group, cite as testbed precedent

- Euler, Lin, Tejedor, Obregon, "High-Altitude Platform Stations as IMT Base Stations (HIBS)" (2022)
- Benzaghta, Geraci, Nikbakht, Lopez-Perez, "UAV Communications in Integrated Terrestrial and Non-terrestrial Networks" (2022)
- Feng, Wang, Chen, Ge, Wang, "Structured Satellite-UAV-Terrestrial Networks for 6G IoT" (2024)

4g. Security in satellite/NTN (6 papers)

- Oligeri, Sciancalepore, Di Pietro, "Physical-layer data of IRIDIUM satellites broadcast messages" (2023)
- Oligeri, Sciancalepore, Sadighian, "SatPrint: Satellite Link Fingerprinting" (2024)
- Zhang, Dai, Ryan, Levin, Heninger, Schulman, "Don't Look Up: Sensitive Internal Links in the Clear on GEO Satellites" (2025, ACM CCS) — high-impact recent security measurement paper

4h. PHY/MAC & 3GPP-standard-adjacent technical papers

(Re-sorted from the auto-classifier — these are core NTN-standard technical contributions, cite heavily as they define the protocol behavior your simulator must reproduce)

- Guidotti, Vanelli-Coralli, Mengali, Cioni, "Non-Terrestrial Networks: Link Budget Analysis" (2020)
 - Lin, Lin, Lowenmark, Rune, Karlsson, "Doppler Shift Estimation in 5G New Radio Non-Terrestrial Networks" (2021)
 - Khan, Lin, "Random Access Preamble Design for 3GPP Non-terrestrial Networks" (2021)
 - Sedin, Feltrin, Lin, "Throughput and Capacity Evaluation of 5G NR NTN with LEO Satellites" (2020)
 - El Jaafari, Chuberre, Anjuere, Combelles, "Introduction to the 3GPP-defined NTN standard" (2023)
 - Kyung, Kim, "Development of Terminal Modem Software for Terrestrial/Non-Terrestrial Communication Based on 3GPP 5G Standards" (2022, x2 near-duplicates)
 - Medina-Acosta, Khan, Mungara, Löwenmark, Tan, Lin, "3GPP Release-19 Physical Layer Enhancements for NR NTN and NB-IoT NTN" (2025) — **most current standards-tracking paper in your library**
 - Prasad, Chandrika, Lampe, Vos, "Enhanced HARQ Scheduling for Non-Terrestrial IoT Networks" (2023)
 - Dureppagari, Saha, Dhillon, Buehrer, "NTN-Based 6G Localization: Vision, Role of LEOs, and Open Problems" (2023)
-

5. THEORY / NEW PERSPECTIVE

- Lu, Yang, Pappas, Geraci, Ma, Quek, "Analysis of Age of Information in Non-terrestrial Networks" (2023, GLOBECOM Wkshp) — stochastic-geometry AoI closed-form model
- Badia, Munari, "Discounted Age of Information for Networks of Constrained Devices" (2022)
- Li, Wu, Kang, Chen, Qiu, Zhou, "A Flexible Topology Control Strategy for Mega-Constellations" (2024) — new "3+1" dynamic-topology conceptual framing
- Hauri et al., "Internet from Space without Inter-satellite Links?" (2020) — reframes a design assumption (ISL necessity) via measurement + analysis
- Kastrati, Badia, "Age of Information in Satellite Networks" (undated) — theoretical framing paper

6. AUTHOR IMPORTANCE BUCKETS

Based on frequency in your library, centrality to NTN standardization, and direct relevance to simulator-building/system-level evaluation.

Most relevant (cite repeatedly; likely reviewers or direct competitors)

Author	Affiliation (typical)	Why critical
Alessandro Vanelli-Coralli (14 entries)	University of Bologna / DLR-adjacent	Leading NTN-standardization voice; co-authors most of the key 3GPP-survey papers you must cite
Alessandro Guidotti (10)	University of Bologna	Co-author on nearly every Vanelli-Coralli NTN architecture/standardization paper; beamforming lead
Andreas Knopp (13)	Bundeswehr University Munich (TUM-adjacent)	Leads the OpenAirInterface 5G-NTN field-trial + satellite swarm/beamforming line — direct testbed competitor to your simulator angle
Tomaso De Cola (11)	DLR (German Aerospace Center)	Broadband SatCom architecture, VNF/emulation platform author — direct simulator-adjacent competitor (Muro2024)
Giovanni Nardini / Giovanni Stea / Antonio Virdis	University of Pisa	Creators of Simu5G itself — the foundational tool you build on; must cite as primary tool reference, not just background

Author	Affiliation (typical)	Why critical
Frank H.P. Fitzek / Jiajing Zhang	TU Dresden	Authors of LEON , your closest direct competitor simulator
Symeon Chatzinotas	University of Luxembourg (SnT)	Prolific NTN PHY/security researcher, appears across beamforming + OAI-NTN experimental papers
Karthikeyan Sundaresan / Sampath Rangarajan / Eugene Chai	NEC Labs America	SkyRAN/SkyLiTE/SkyHaul/SkyCore — the closest systems-building precedent (UAV-LTE) for your "build and release a working network stack" contribution narrative

Important (cite 1-2 papers each; establish sub-theme credibility)

Giuseppe Araniti, Antonella Molinaro, Sara Pizzi, Federica Rinaldi (Univ. Mediterranea Reggio Calabria — NTN service/mobility group) · Xingqin Lin (Nvidia/Ericsson-adjacent, 3GPP NTN standardization) · Zhi Lin / Min Lin (secure beamforming, satellite-terrestrial coexistence) · Enrico Del Re / Romano Fantacci / Giovanni Giambene (legacy LEO-MSS handover theory, still-cited foundational work) · Mohamed El Jaafari / Nicolas Chuberre (Thales Alenia Space — 3GPP NTN rapporteurs) · Daniele Pugliese / Domenico Striccoli / Luigi Alfredo Grieco (Politecnico di Bari — IAB system-level simulator, your closest architecture-level competitor) · Ju Hyung Lee / Young Chai Ko (DRL-based handover) · Kanglian Zhao / Wenfeng Li (satellite-terrestrial edge computing/routing, China) · Cedric Westphal / Lin Han / Richard Li (Futurewei — IETF/3GPP satellite-networking surveys) · Wolfgang Kellerer / Jorg Von Mankowski (TU Munich — aeronautical MEC/SDN)

Okay (cite once if needed for a specific claim; niche/single-topic)

Diego Tuzi, Thomas Delamotte (D2C swarm antenna arrays) · Giovanni Geraci, David Lopez-Perez (3D TN-NTN opportunities overview) · Mustafa Ozger, Cicek Cavdar (HAPS reliability analysis) · Leonardo Goratti, Konstantinos Liolis (SES/testbed integration) · Andrea Munari, Federico Clazzer (DLR — random access/AoI theory) · Peter Ashwood-Smith (orthodromic routing, Huawei) · Gabriele Oliveri, Savio Sciancalepore (satellite security/fingerprinting, HBKU) · Wanshi Chen, Antti Toskala (5G-Advanced/6G roadmap, Qualcomm/Nokia) · Harpreet Dhillon, R. Michael Buehrer (NTN localization theory, Virginia Tech)

Not important (legacy/tangential — cite at most as one historical footnote, or drop)

- All Iridium/Globalstar/Inmarsat *system description* papers pre-2015 (Crowe 1999, Comparetto 1994, Finean 1996, Maine/Devieux/Swan 1995, Helm 2009, Denney 2007, Jabbar 2004, Wang/Liu 2011-INMARSAT) — pure historical background, cite **at most one** sentence citing 1-2 of these for "satellite comms predates 5G-NTN" framing

- Pulak Chowdhury / Mohammed Atiquzzaman handover survey (2006) — cite once as the classic taxonomy, do not cite the duplicate/garbled re-entries
- UAV-robotics/SLAM papers unrelated to NTN (Mahdoui et al. cooperative SLAM 2020, Ahmad et al. AeroTraj LiDAR 2023, Choi et al. indoor UAV navigation 2021) — **exclude**, off-topic for a TN-NTN simulator paper
- Automotive Ethernet/CAN/FlexRay OMNeT++ paper (Steinbach et al.) — tangential; only relevant if you discuss OMNeT++ DSL tooling generally, otherwise drop
- Malformed/placeholder entries with no real title ("", "`CHALLENGES`", "`HARQ`", "`Key Items`", "`Guest_Editorial`", "`MEC_5G`", "`MEC_satcom`", "`Lifeofpacket`", "`Iridium-Leo`", IEEE Xplore stub PDFs, bare conference-proceedings-only (`@book` entries) — **delete from BibTeX before submission**, these are reference-manager import artifacts, not real citable sources

7. STANDARDS-BODY CONTRIBUTIONS (3GPP / ETSI / ITU-R)

Your library contains only secondary literature *about* 3GPP standards (El Jaafari 2023, Masini 2023, Kyung/Kim 2022, Medina-Acosta 2025, TR 121 917) — you should cite the **primary standards documents themselves** directly, as an IEEE Access reviewer will expect this. Recommended primary references to add:

3GPP (Release 15-19, NTN-relevant)

Document	Title	Relevance
3GPP TR 38.811	Study on New Radio (NR) to support Non-Terrestrial Networks (Rel-15)	Foundational NTN channel model & scenario definitions — cite for your channel-model implementation
3GPP TR 38.821	Solutions for NR to support Non-Terrestrial Networks (Rel-16)	Architecture options, timing advance, HARQ/RRC adaptations — core reference for protocol-stack modeling
3GPP TS 38.300 / 38.331	NR RRC protocol specification, with NTN amendments (Rel-17)	Direct spec you must reproduce in your RRC/mobility-management module
3GPP TR 38.863 / TS 38.101-5	NTN UE RF requirements	If you model UE-side link budget
3GPP TS 22.261	Service requirements for the 5G system, incl. NTN service continuity	Service-continuity requirements motivating your TN ↔ NTN switchover logic

Document	Title	Relevance
3GPP TR 22.822	Study on using satellite access in 5G	Rel-16 use-case study — good Background-section citation
3GPP TR 38.821 Rel-17 updates / Rel-18 "5G-Advanced NTN" WID, Rel-19 NTN enhancements	Ongoing NTN work items (regenerative payload, NB-IoT NTN, mobile IoT NTN)	Cite for "current/ongoing standardization" framing — pairs with Medina-Acosta 2025 already in your library
3GPP TR 21.917	Summary of Rel-17 Work Items (already in your .bib)	Good general Rel-17 anchor citation

ETSI

Document	Title	Relevance
ETSI TR 103 611	Satellite Earth Stations and Systems (SES); Overview of methods for 5G NTN	ETSI's own NTN overview, complements 3GPP view
ETSI GS NFV series	NFV architectural framework	If you position VNF/orchestration aspects (relevant given Muro2024 in your library)
ETSI EN 302 307 (DVB-S2/S2X)	Second-generation framing structure for satellite broadcasting	Relevant if your PHY layer references DVB waveforms for GEO backhaul comparisons

ITU-R

Document	Title	Relevance
ITU-R Recommendations on IMT-2020 (5G) non-terrestrial component	Spectrum/coexistence framework	High-level regulatory context for NTN spectrum sharing
ITU-R M.2514 / related IMT-2020 satellite reports	Technical performance requirements incl. satellite	Useful one-line citation for global regulatory framing

IETF (not a formal SDO for RAN, but relevant to your routing/addressing discussion)

- IETF Satellite Networking working-group drafts (referenced in

Han/Retana/Westphal/Li papers already in your library — cite those as secondary sources, and optionally add the specific IETF drafts on LEO addressing/routing if you discuss network-layer routing).

8. Suggested Background-section paragraph structure

1. **Para 1 — Why NTN matters now:** cite 2–3 Overview papers (Guidotti 2020 survey, Geraci 2023, Wang 2026)
 2. **Para 2 — 3GPP/ETSI standardization trajectory:** cite El Jaafari 2023, Masini 2023, Medina-Acosta 2025 + primary 3GPP TR 38.811/38.821/38.300 docs
 3. **Para 3 — Technical breadth of NTN research (handover, beamforming, routing, edge computing):** cite 1–2 representative papers per sub-theme from Section 4 above — this establishes "the field is active but fragmented"
 4. **Para 4 — Existing simulation/testbed tools and their limitations:** walk through Section 2 table (Simu5G → LEON → Pugliese IAB sim → disaster-aware TN-NTN sim → OAI testbeds → Jiang et al. survey) ending in your explicit gap statement
 5. **Para 5 — Your contribution:** state the OMNeT++/Simu5G/INET + SUMO/VEINS heterogeneous-UE, open-source, academia-oriented TN-NTN simulator as filling the identified gap, explicitly contrasting with LEON (no SUMO/VEINS, handover-only) and the disaster-aware simulator (no heterogeneous UE mobility modeling)
-

9. Immediate action items for you

1. **Add to Zotero/BibTeX:** Jiang et al. 2023 IEEE Access survey; LEON's arXiv/ACM full citation (already have it, but verify DOI); the Disaster-Aware TN-NTN simulator (arXiv 2605.05852); Pugliese 2024 (already have, verify not duplicate with the IAB architectural-analysis undated entry earlier in your file — you have **two Pugliese/Striccoli entries that look like the same paper at different stages**, consolidate); 5GSimV2IN GitHub citation; primary 3GPP TR 38.811/38.821/38.300 and ETSI TR 103 611.
2. **De-duplicate:** your .bib has several exact or near-duplicate entries (e.g., two "Trends in LEO Satellite Handover Algorithms" by Park/Kim; two "Pre-Coded Inter-Satellite Routing Algorithm..." by Zhang et al.; two Simu5G papers by Nardini et al. — keep one canonical citation each; two "Development of Terminal Modem Software..." by Kyung/Kim; two Chowdhury/Atiquzzaman handover surveys; two "Guest Editorial Advances in Satellite Communications" Part 1/2 — keep both, they're genuinely different parts).
3. **Delete the ~15 malformed/blank-title entries** listed in Section 6 "Not important."
4. **Verify the "1970" and "133" year entries** (Riad/Elsokkary routing paper, Berg Iridium Certus report) — likely BibTeX year-field parsing errors from the original export; fix before submission.

